

Medical Patient Data Analysis Using R

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1 Introduction

This project demonstrates basic medical patient data analysis using R. The dataset contains patient age, gender, blood pressure, cholesterol, glucose level, and health status.

Objectives: - Import and clean data - Perform statistical analysis - Create different graphs - Interpret results - Present findings and conclusion

2 1. Data Importing

```
medical_data <- data.frame(  
  Patient_ID = 101:115,  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female", "Male"),  
  Age = c(25, 32, 45, 29, 54, 41, 36, 60, 50, 27, 48, 35, 65, 52, 39),  
  Blood_Pressure = c(118, 122, 135, 115, 145, 130, 125, 150, 140, 110, 138, 120, 155, 142, 128),  
  Cholesterol = c(180, 190, 220, 175, 245, 210, 195, 260, 235, 165, 225, 185, 270, 240, 205),  
  Glucose = c(90, 95, 110, 88, 125, 115, 100, 135, 120, 85, 118, 92, 140, 130, 105),  
  Health_Status = c("Normal", "Normal", "At Risk", "Normal", "At Risk", "At Risk", "Normal", "At Risk", "At Risk", "Normal", "At Risk", "Normal", "At Risk", "Normal", "At Risk"),  
)  
  
write.csv(medical_data, "medical_data.csv", row.names=FALSE)  
medical_data <- read.csv("medical_data.csv")  
head(medical_data)
```

##	Patient_ID	Gender	Age	Blood_Pressure	Cholesterol	Glucose	Health_Status
## 1	101	Male	25	118	180	90	Normal
## 2	102	Female	32	122	190	95	Normal
## 3	103	Male	45	135	220	110	At Risk
## 4	104	Female	29	115	175	88	Normal
## 5	105	Male	54	145	245	125	At Risk
## 6	106	Female	41	130	210	115	At Risk

Explanation: `data.frame()` creates data, `write.csv()` saves it, `read.csv()` imports it, and `head()` displays records.

3 2. Data Preprocessing and Cleaning

```
str(medical_data)
```

```
## 'data.frame':    15 obs. of  7 variables:
## $ Patient_ID    : int  101 102 103 104 105 106 107 108 109 110 ...
## $ Gender        : chr  "Male" "Female" "Male" "Female" ...
## $ Age           : int  25 32 45 29 54 41 36 60 50 27 ...
## $ Blood_Pressure: int  118 122 135 115 145 130 125 150 140 110 ...
## $ Cholesterol   : int  180 190 220 175 245 210 195 260 235 165 ...
## $ Glucose       : int  90 95 110 88 125 115 100 135 120 85 ...
## $ Health_Status : chr  "Normal" "Normal" "At Risk" "Normal" ...
```

```
colSums(is.na(medical_data))
```

```
##      Patient_ID      Gender      Age Blood_Pressure      Cholesterol
##           0           0           0           0           0
##      Glucose Health_Status
##           0           0
```

```
sum(duplicated(medical_data))
```

```
## [1] 0
```

```
summary(medical_data)
```

```
##      Patient_ID      Gender      Age      Blood_Pressure      Cholesterol
##  Min.    :101.0    Length    :15    Min.    :25.00    Min.    :110.0    Min.    :165.0
##  1st Qu.:104.5    N.unique  : 2    1st Qu.:33.50    1st Qu.:121.0    1st Qu.:187.5
##  Median :108.0    N.blank   : 0    Median :41.00    Median :130.0    Median :210.0
##  Mean   :108.0    Min.nchar: 4    Mean   :42.53    Mean   :131.5    Mean   :213.3
##  3rd Qu.:111.5    Max.nchar: 6    3rd Qu.:51.00    3rd Qu.:141.0    3rd Qu.:237.5
##  Max.    :115.0                      Max.    :65.00    Max.    :155.0    Max.    :270.0
##      Glucose      Health_Status
##  Min.    : 85.0    Length    :15
##  1st Qu.: 93.5    N.unique  : 2
##  Median :110.0    N.blank   : 0
##  Mean   :109.9    Min.nchar: 6
##  3rd Qu.:122.5    Max.nchar: 7
##  Max.    :140.0
```

Interpretation: The data is checked for structure, missing values, duplicates, and basic statistics before analysis.

4 3. Statistical Analysis

```
mean(medical_data$Age)
```

```
## [1] 42.53333
```

```
median(medical_data$Age)
```

```
## [1] 41
```

```
var(medical_data$Age)
```

```
## [1] 148.5524
```

```
sd(medical_data$Age)
```

```
## [1] 12.18821
```

```
mean(medical_data$Blood_Pressure)
```

```
## [1] 131.5333
```

```
median(medical_data$Blood_Pressure)
```

```
## [1] 130
```

```
sd(medical_data$Blood_Pressure)
```

```
## [1] 13.38905
```

```
mean(medical_data$Cholesterol)
```

```
## [1] 213.3333
```

```
median(medical_data$Cholesterol)
```

```
## [1] 210
```

```
sd(medical_data$Cholesterol)
```

```
## [1] 32.10845
```

```
mean(medical_data$Glucose)
```

```
## [1] 109.8667
```

```
median(medical_data$Glucose)
```

```
## [1] 110
```

```
sd(medical_data$Glucose)
```

```
## [1] 17.97167
```

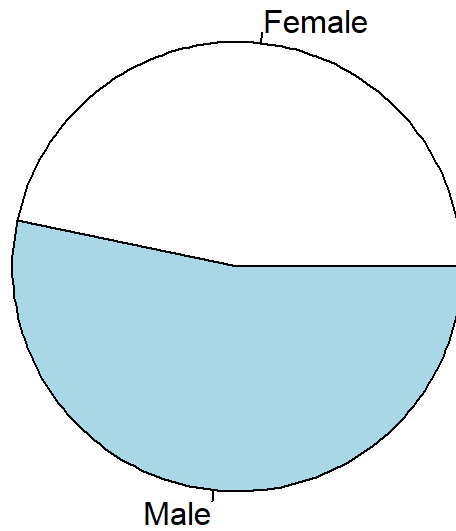
Explanation: Mean and median show typical values. Variance and standard deviation show variation in numerical data.

5 4. Graphical Analysis

5.1 4.1 Gender Distribution

```
pie(table(medical_data$Gender),  
     main="Gender Distribution")
```

Gender Distribution

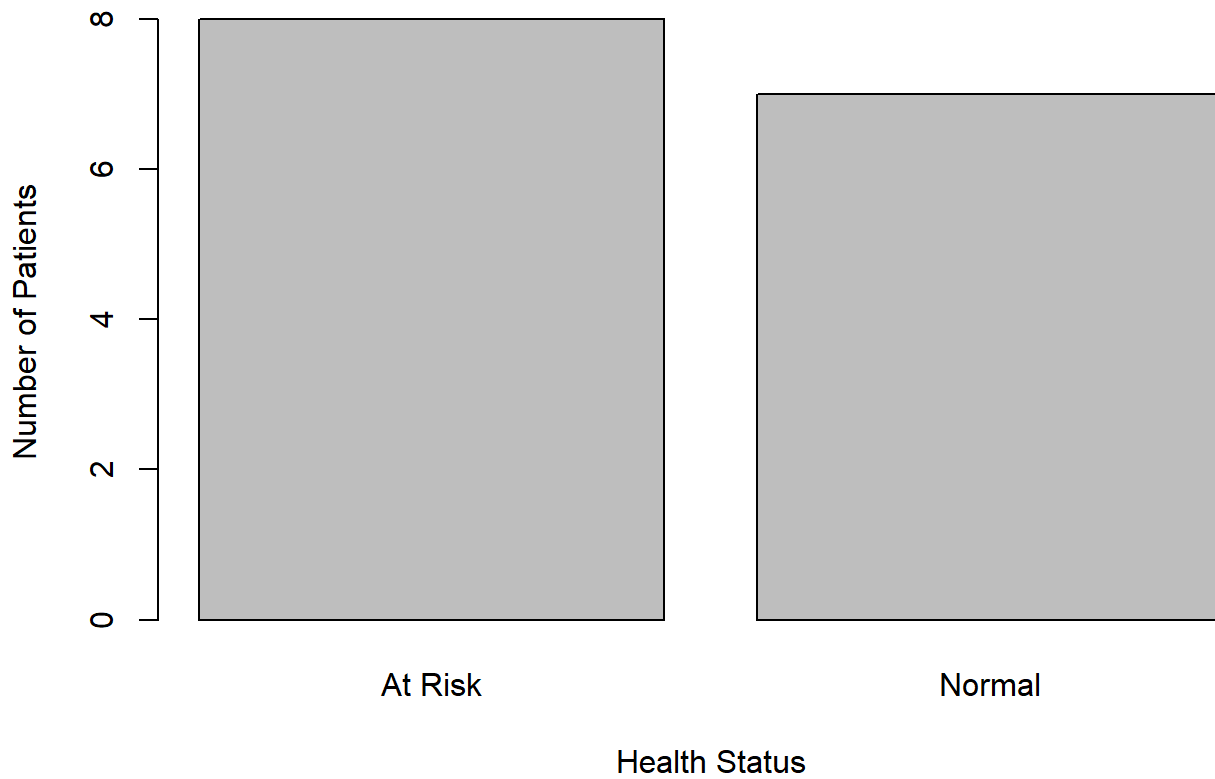


Interpretation: Shows the proportion of male and female patients.

5.2 4.2 Health Status

```
barplot(table(medical_data$Health_Status),  
        main="Health Status Distribution",  
        xlab="Health Status",  
        ylab="Number of Patients")
```

Health Status Distribution

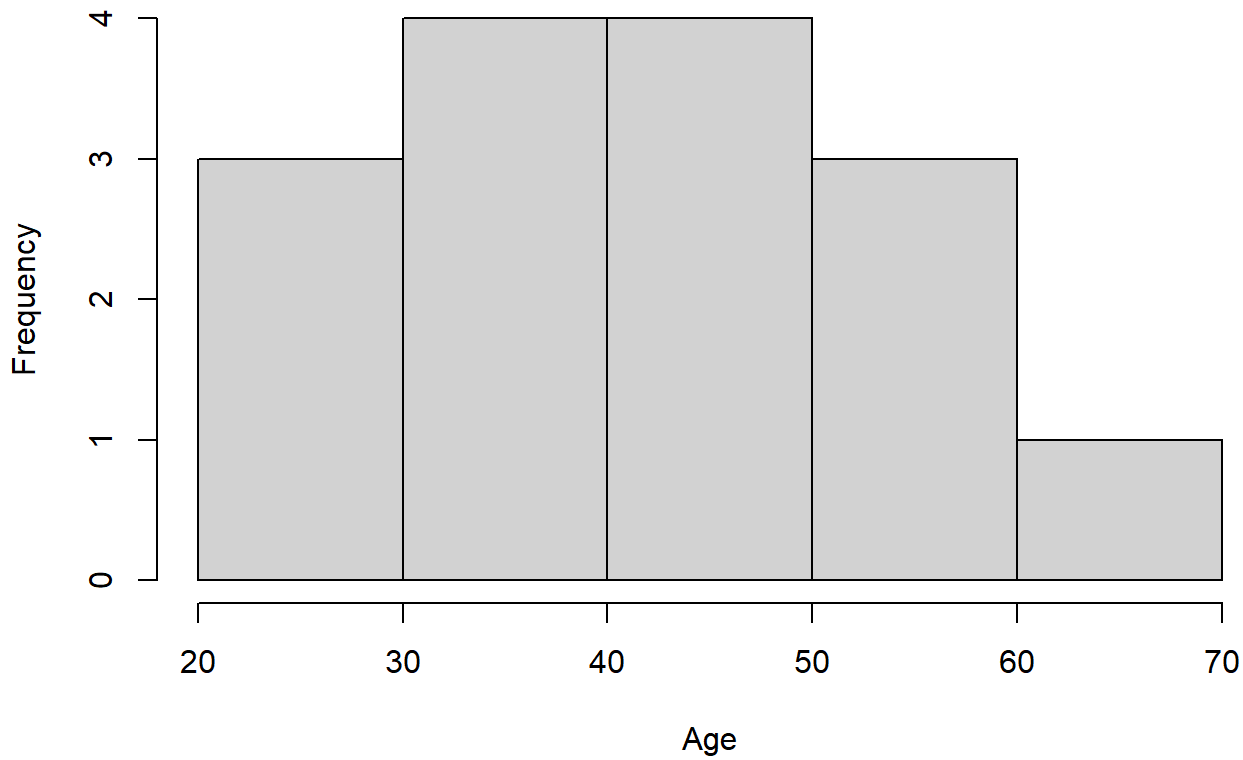


Interpretation: Compares the number of patients in each health-status category.

5.3 4.3 Age Distribution

```
hist(medical_data$Age,  
     main="Age Distribution",  
     xlab="Age",  
     ylab="Frequency")
```

Age Distribution

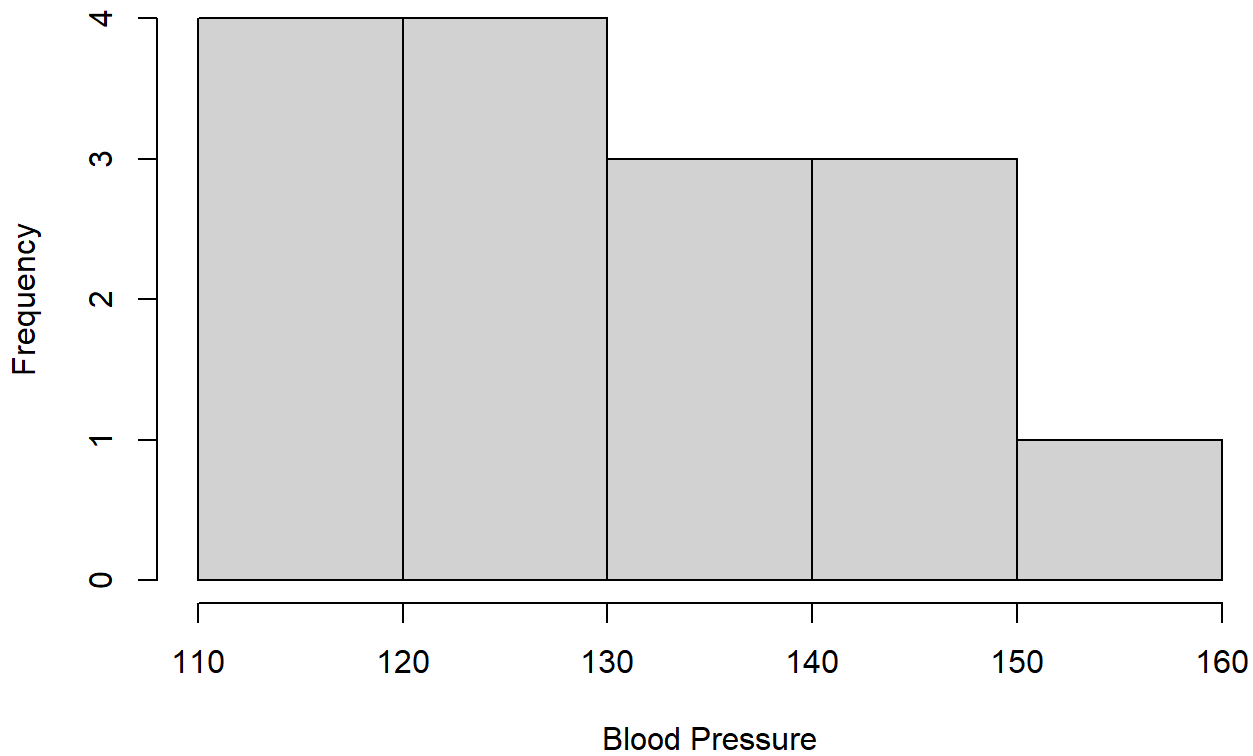


Interpretation: Shows how patient ages are distributed.

5.4 4.4 Blood Pressure Distribution

```
hist(medical_data$Blood_Pressure,  
     main="Blood Pressure Distribution",  
     xlab="Blood Pressure",  
     ylab="Frequency")
```

Blood Pressure Distribution

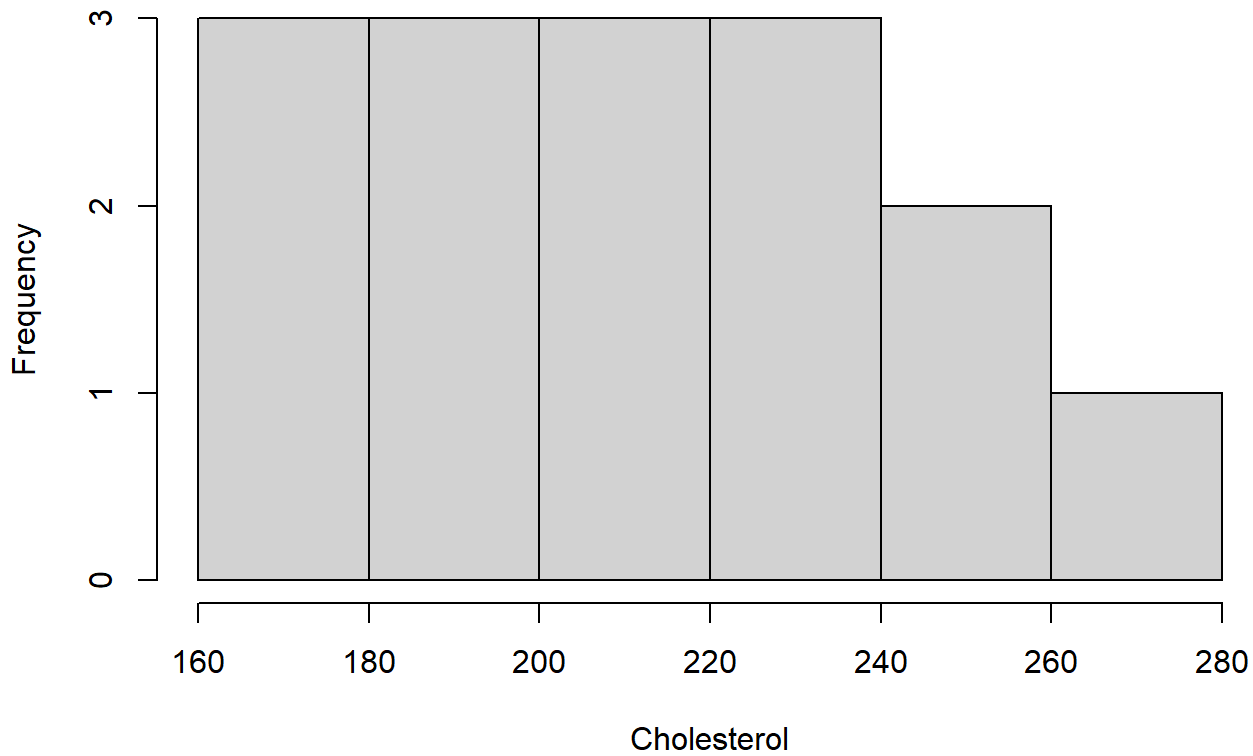


Interpretation: Shows the distribution of blood pressure measurements.

5.5 4.5 Cholesterol Distribution

```
hist(medical_data$Cholesterol,  
     main="Cholesterol Distribution",  
     xlab="Cholesterol",  
     ylab="Frequency")
```

Cholesterol Distribution

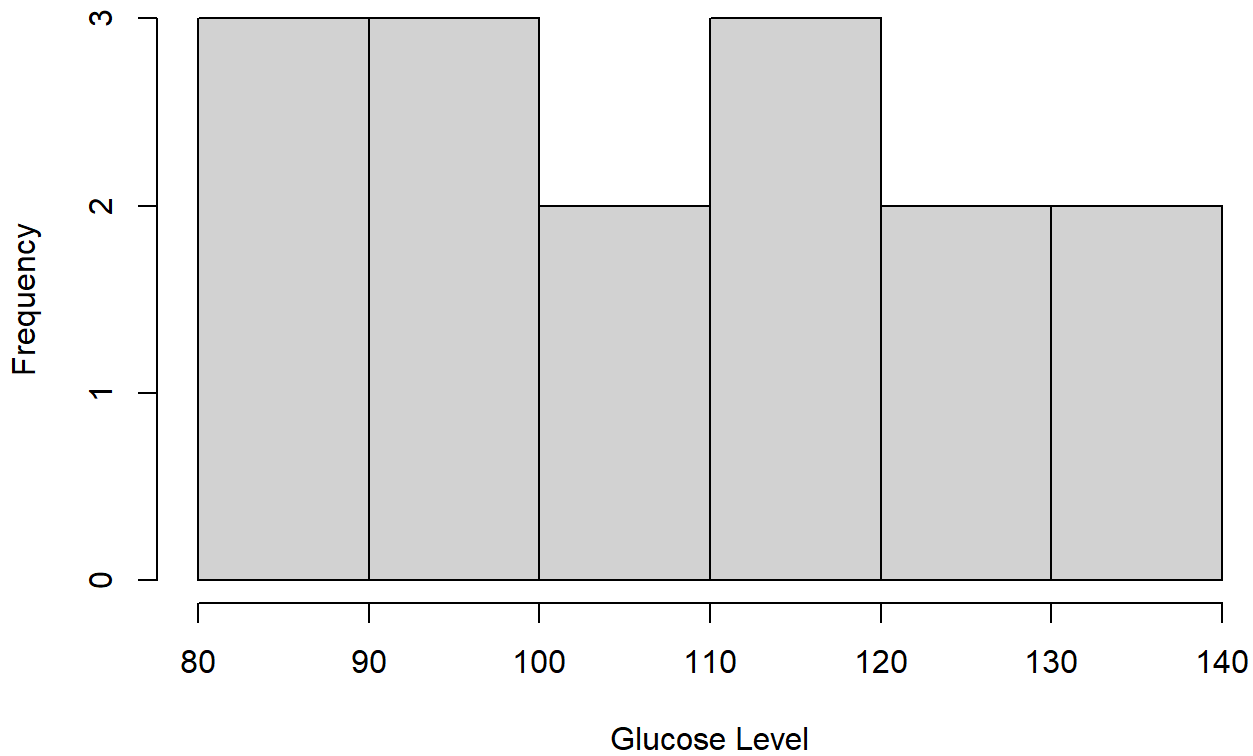


Interpretation: Shows the spread of cholesterol measurements.

5.6 4.6 Glucose Distribution

```
hist(medical_data$Glucose,  
     main="Glucose Level Distribution",  
     xlab="Glucose Level",  
     ylab="Frequency")
```

Glucose Level Distribution

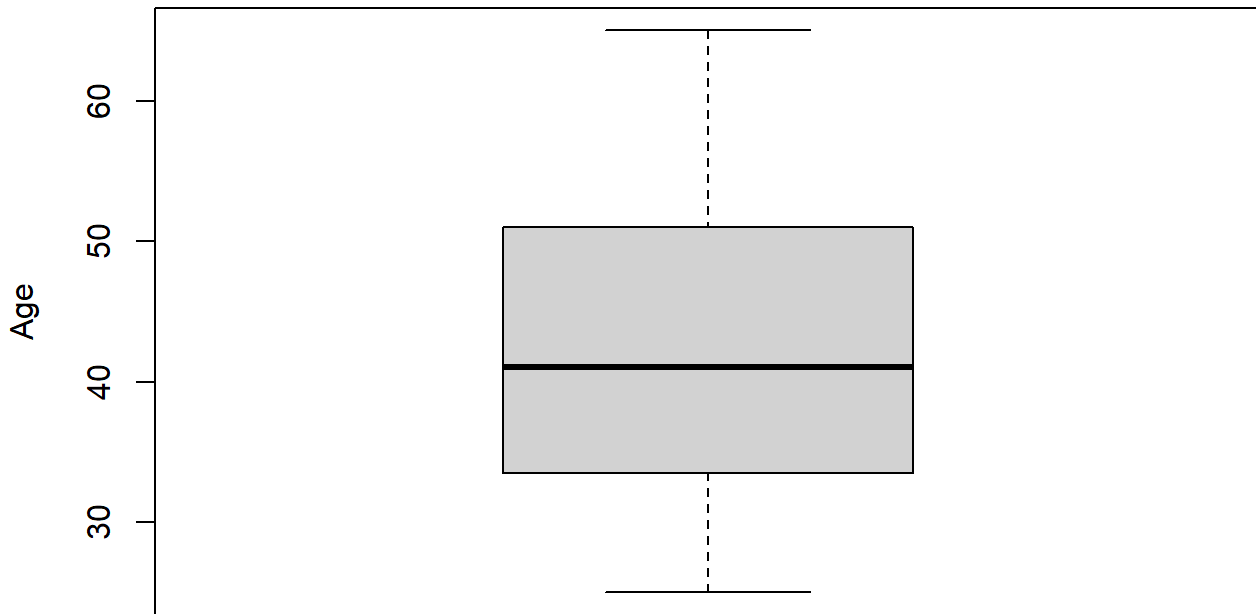


Interpretation: Shows the distribution of glucose measurements.

5.7 4.7 Age Boxplot

```
boxplot(medical_data$Age,  
        main="Patient Age Boxplot",  
        ylab="Age")
```

Patient Age Boxplot

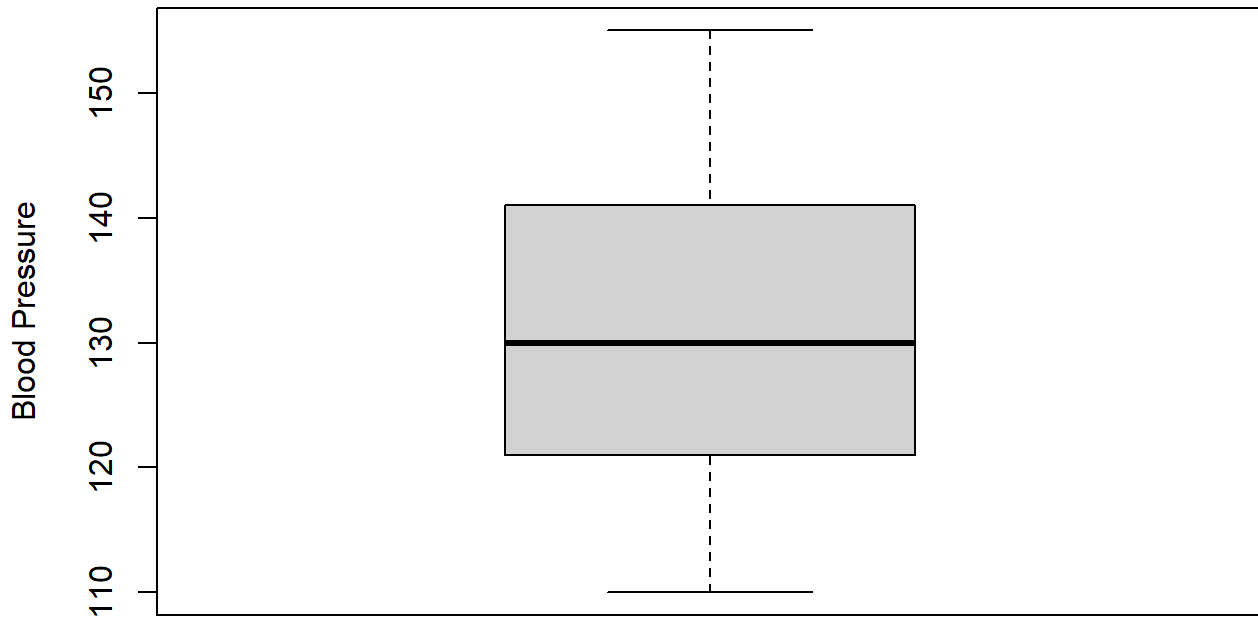


Interpretation: Shows median, spread, and possible unusual age values.

5.8 4.8 Blood Pressure Boxplot

```
boxplot(medical_data$Blood_Pressure,  
        main="Blood Pressure Boxplot",  
        ylab="Blood Pressure")
```

Blood Pressure Boxplot

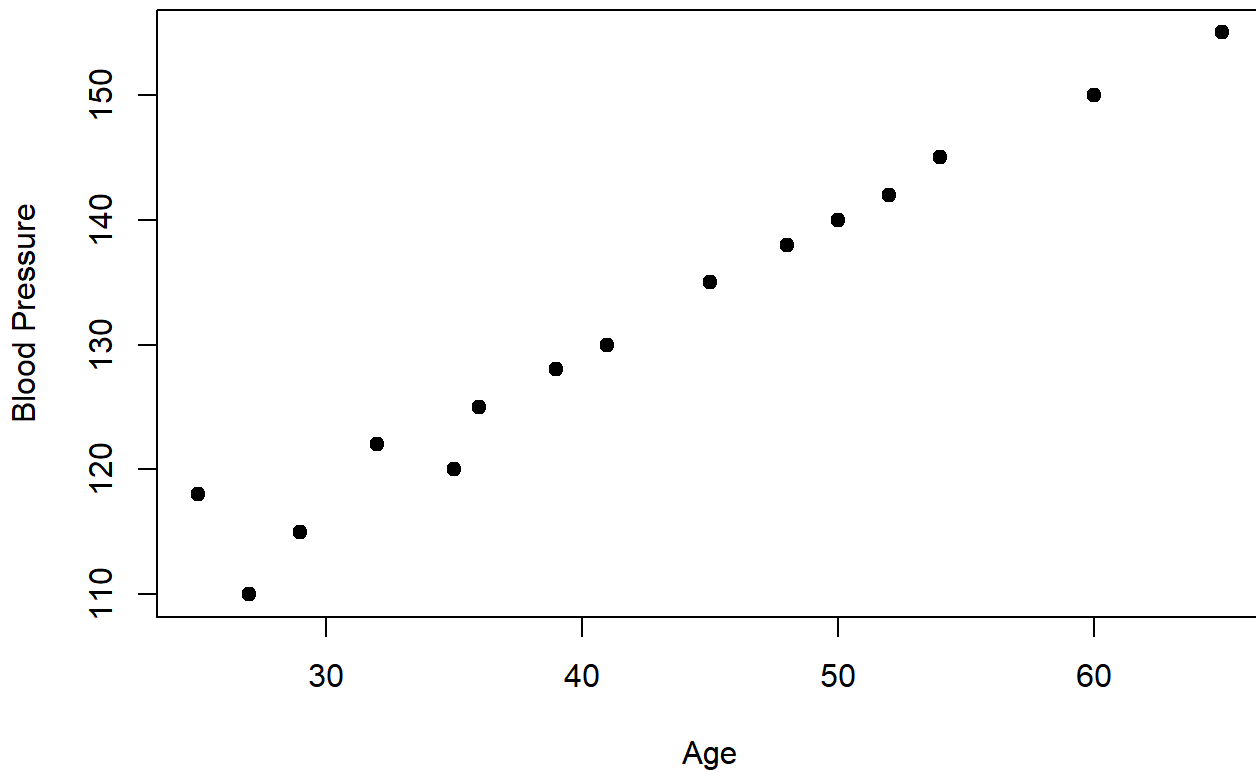


Interpretation: Shows the spread of blood pressure values.

5.9 4.9 Age vs Blood Pressure

```
plot(medical_data$Age, medical_data$Blood_Pressure,  
     main="Age vs Blood Pressure",  
     xlab="Age", ylab="Blood Pressure", pch=19)
```

Age vs Blood Pressure

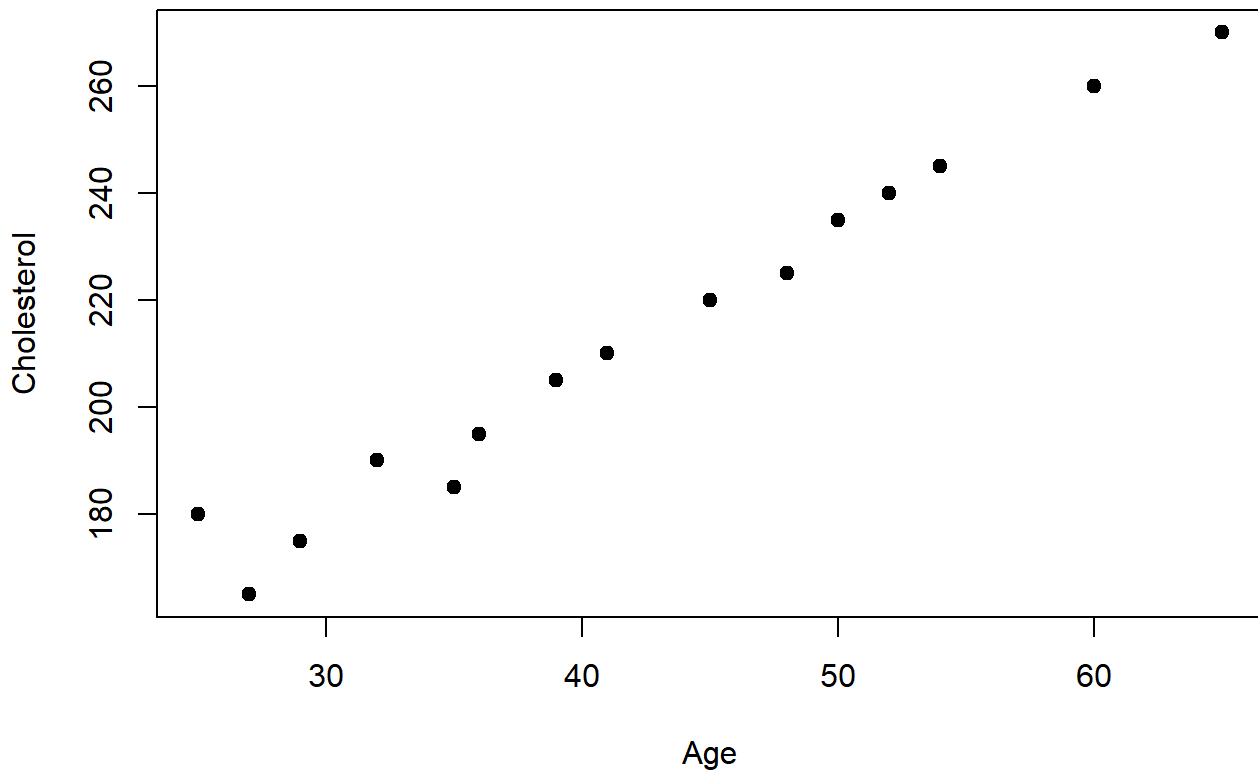


Interpretation: Helps visually explore the relationship between age and blood pressure in this sample; it is not proof of a medical relationship.

5.10 4.10 Age vs Cholesterol

```
plot(medical_data$Age, medical_data$Cholesterol,  
     main="Age vs Cholesterol",  
     xlab="Age", ylab="Cholesterol", pch=19)
```

Age vs Cholesterol

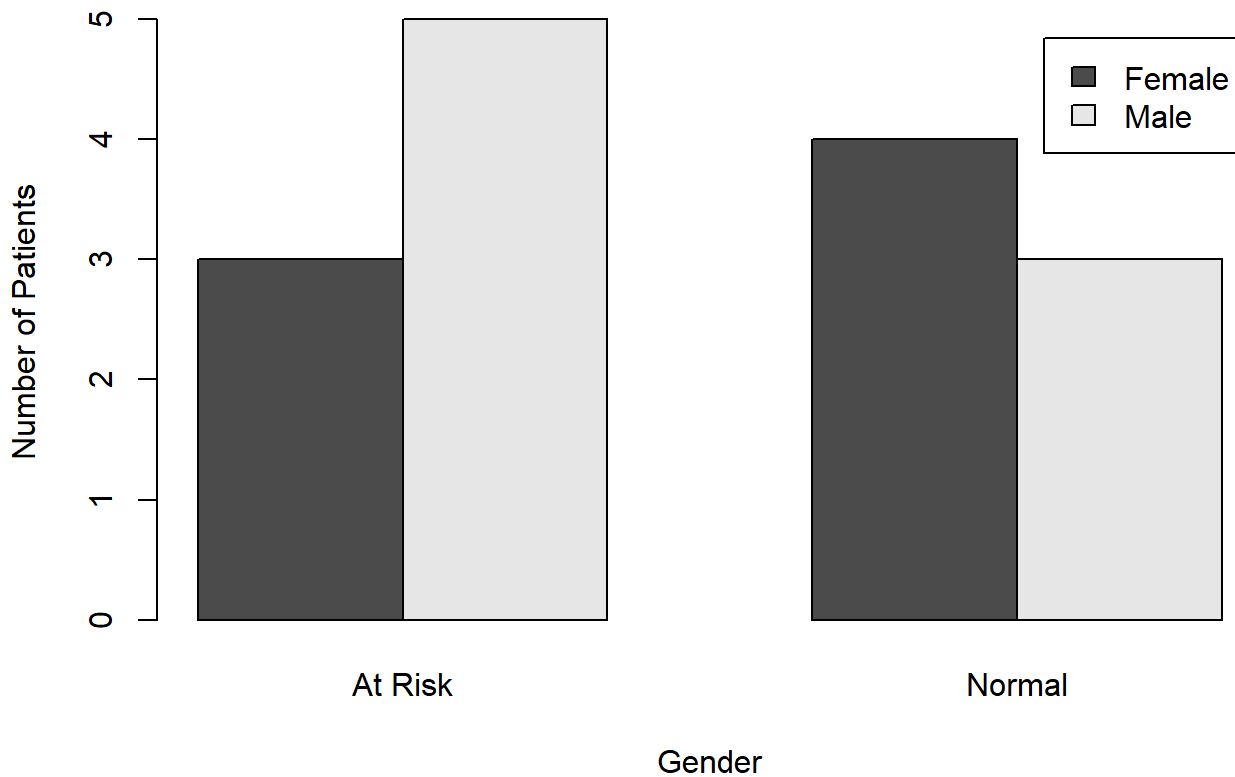


Interpretation: Shows how cholesterol measurements are distributed across patient ages.

5.11 4.11 Gender-wise Health Status

```
barplot(table(medical_data$Gender, medical_data$Health_Status),
        beside=TRUE,
        legend.text=TRUE,
        main="Health Status by Gender",
        xlab="Gender",
        ylab="Number of Patients")
```

Health Status by Gender



Interpretation: Compares health-status categories across genders.

6 5. Findings

1. The dataset contains 15 patient records.
2. Patient age, gender, blood pressure, cholesterol, glucose, and health status were analyzed.
3. Missing and duplicate values were checked.
4. Mean, median, variance, and standard deviation were calculated.
5. Multiple graphs were used to understand categorical and numerical data.
6. Histograms show distributions, boxplots show spread, and scatter plots help explore relationships.
7. The analysis is descriptive and based only on the sample dataset.

7 6. Conclusion

This project demonstrates how R can be used for basic medical patient data analysis. The data was imported, cleaned, statistically analyzed, and visualized using different graphs. R provides simple functions for data analysis and visualization. The results are for academic demonstration only and should not be considered medical diagnosis or clinical advice.

8 7. Complete Code

```
medical_data <- data.frame(
  Patient_ID=101:115,
  Gender=c("Male","Female","Male","Female","Male","Female","Male","Female","Male","Female","Male",
"Female","Male","Female","Male"),
  Age=c(25,32,45,29,54,41,36,60,50,27,48,35,65,52,39),
  Blood_Pressure=c(118,122,135,115,145,130,125,150,140,110,138,120,155,142,128),
  Cholesterol=c(180,190,220,175,245,210,195,260,235,165,225,185,270,240,205),
  Glucose=c(90,95,110,88,125,115,100,135,120,85,118,92,140,130,105),
  Health_Status=c("Normal","Normal","At Risk","Normal","At Risk","At Risk","Normal","At Risk","At Risk",
"Risk","Normal","At Risk","Normal","At Risk","At Risk","Normal")
)

str(medical_data)
```

```
## 'data.frame':    15 obs. of  7 variables:
## $ Patient_ID    : int  101 102 103 104 105 106 107 108 109 110 ...
## $ Gender        : chr  "Male" "Female" "Male" "Female" ...
## $ Age           : num  25 32 45 29 54 41 36 60 50 27 ...
## $ Blood_Pressure: num  118 122 135 115 145 130 125 150 140 110 ...
## $ Cholesterol   : num  180 190 220 175 245 210 195 260 235 165 ...
## $ Glucose       : num  90 95 110 88 125 115 100 135 120 85 ...
## $ Health_Status : chr  "Normal" "Normal" "At Risk" "Normal" ...
```

```
colSums(is.na(medical_data))
```

```
##      Patient_ID      Gender      Age Blood_Pressure      Cholesterol
##           0           0           0           0           0
##      Glucose Health_Status
##           0           0
```

```
sum(duplicated(medical_data))
```

```
## [1] 0
```

```
summary(medical_data)
```

```
##      Patient_ID      Gender      Age      Blood_Pressure      Cholesterol
## Min.   :101.0    Length   :15    Min.   :25.00    Min.   :110.0    Min.   :165.0
## 1st Qu.:104.5    N.unique  : 2    1st Qu.:33.50    1st Qu.:121.0    1st Qu.:187.5
## Median :108.0    N.blank   : 0    Median :41.00    Median :130.0    Median :210.0
## Mean   :108.0    Min.nchar : 4    Mean   :42.53    Mean   :131.5    Mean   :213.3
## 3rd Qu.:111.5    Max.nchar : 6    3rd Qu.:51.00    3rd Qu.:141.0    3rd Qu.:237.5
## Max.    :115.0                Max.    :65.00    Max.    :155.0    Max.    :270.0
##      Glucose      Health_Status
## Min.   : 85.0    Length   :15
## 1st Qu.: 93.5    N.unique  : 2
## Median :110.0    N.blank   : 0
## Mean   :109.9    Min.nchar : 6
## 3rd Qu.:122.5    Max.nchar : 7
## Max.    :140.0
```

```
mean(medical_data$Age)
```

```
## [1] 42.53333
```

```
median(medical_data$Age)
```

```
## [1] 41
```

```
var(medical_data$Age)
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## [1] 148.5524
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sd(medical_data$Age)
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## [1] 12.18821
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mean(medical_data$Blood_Pressure)
```

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## [1] 131.5333
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median(medical_data$Blood_Pressure)
```

```
## [1] 130
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sd(medical_data$Blood_Pressure)
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mean(medical_data$Cholesterol)
```

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## [1] 213.3333
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median(medical_data$Cholesterol)
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## [1] 210
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```
sd(medical_data$Cholesterol)
```

```
## [1] 32.10845
```

```
mean(medical_data$Glucose)
```

```
## [1] 109.8667
```

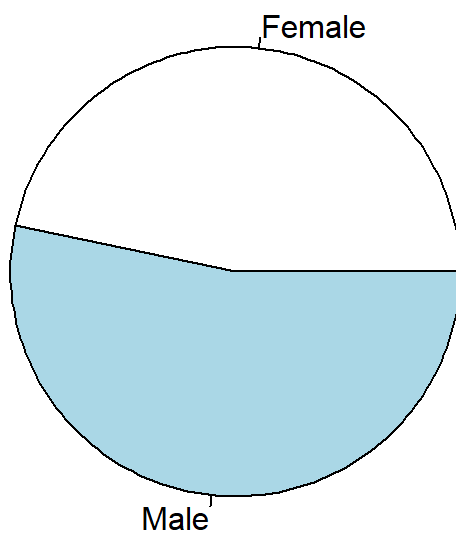
```
median(medical_data$Glucose)
```

```
## [1] 110
```

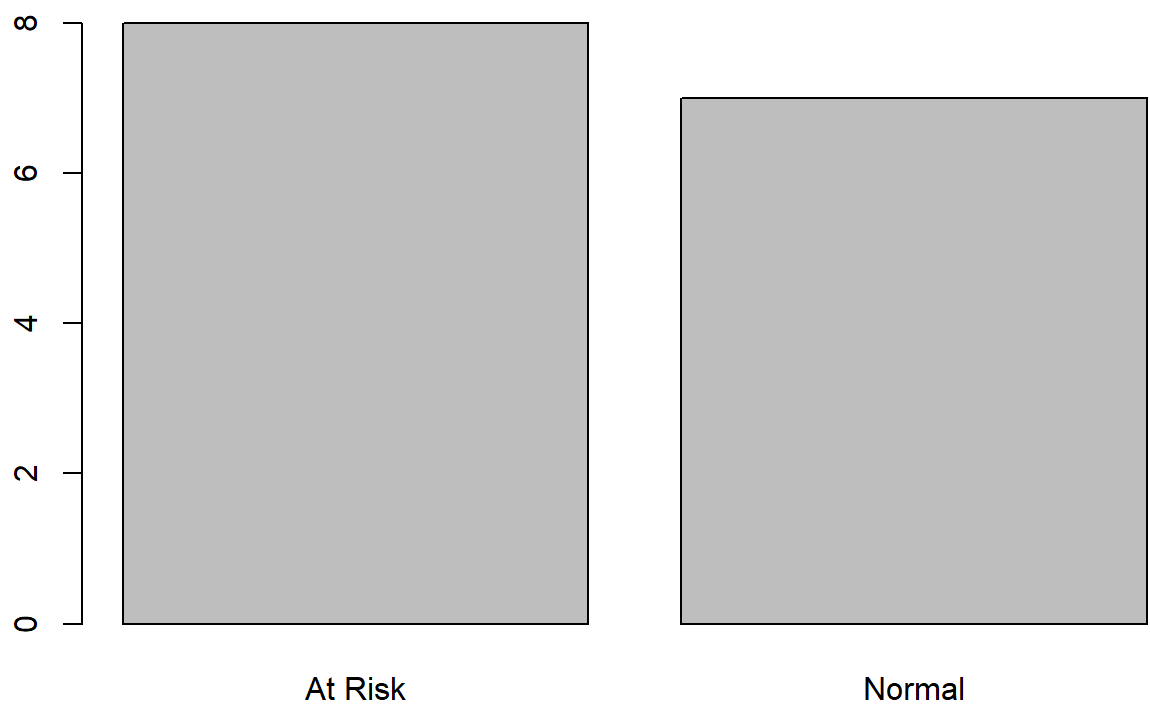
```
sd(medical_data$Glucose)
```

```
## [1] 17.97167
```

```
pie(table(medical_data$Gender))
```

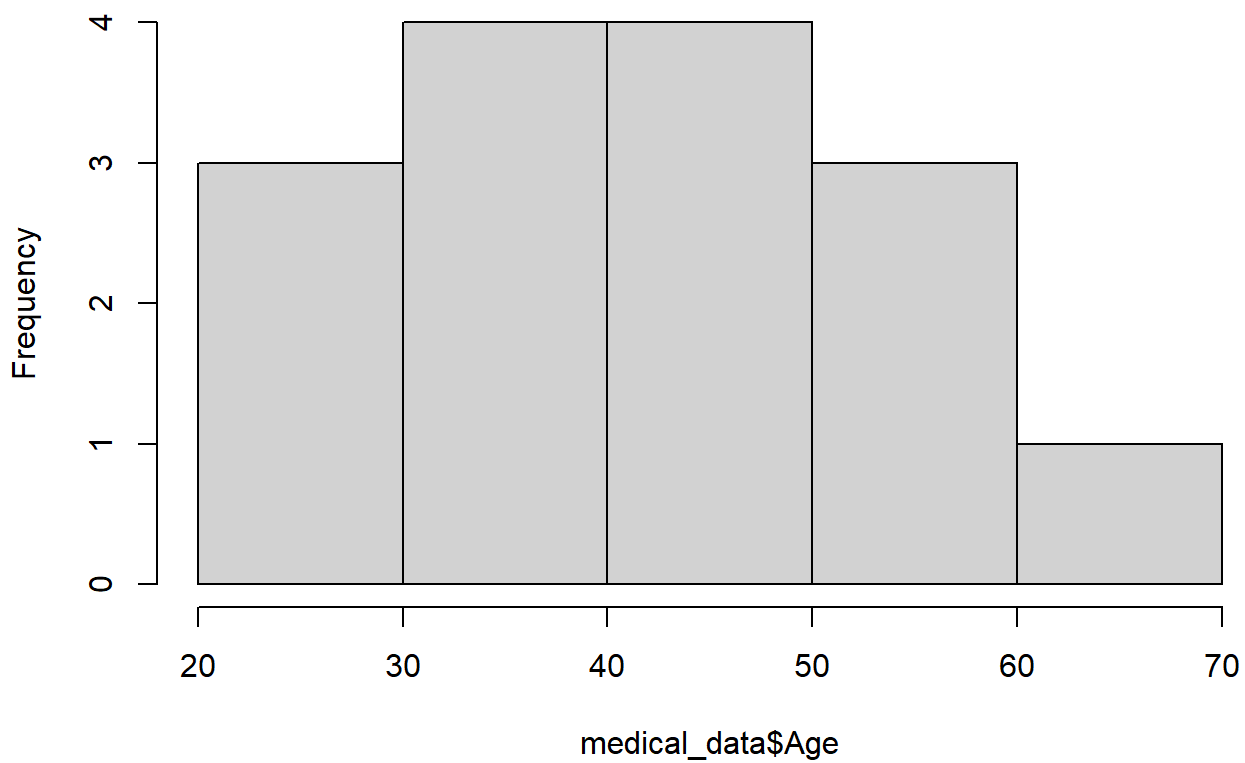


```
barplot(table(medical_data$Health_Status))
```



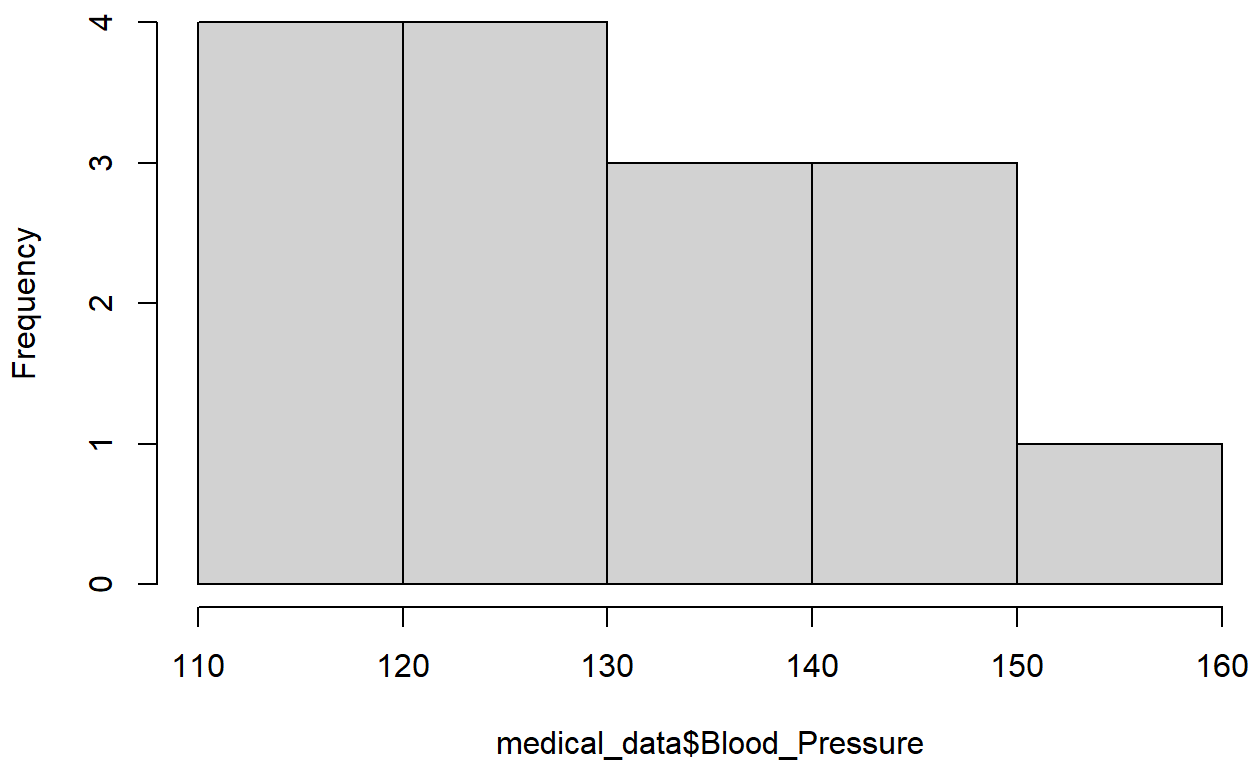
```
hist(medical_data$Age)
```

Histogram of medical_data\$Age



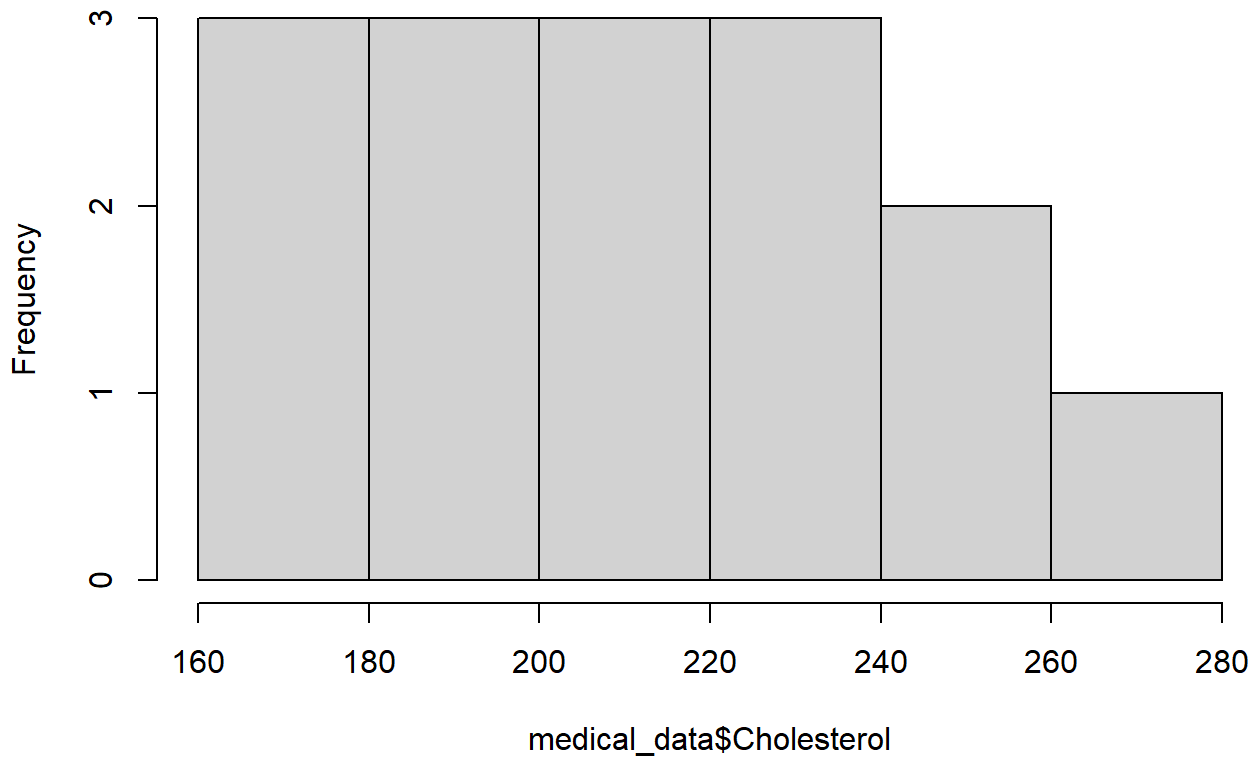
```
hist(medical_data$Blood_Pressure)
```

Histogram of medical_data\$Blood_Pressure



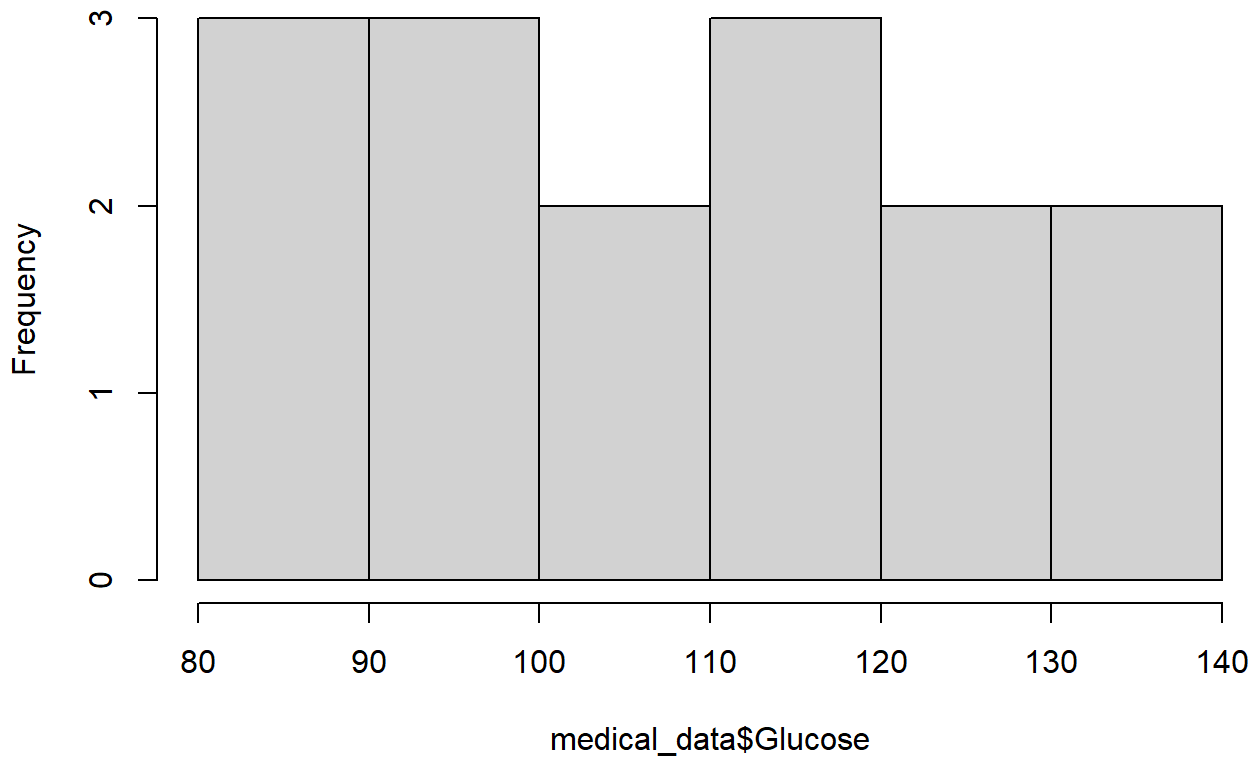
```
hist(medical_data$Cholesterol)
```

Histogram of medical_data\$Cholesterol

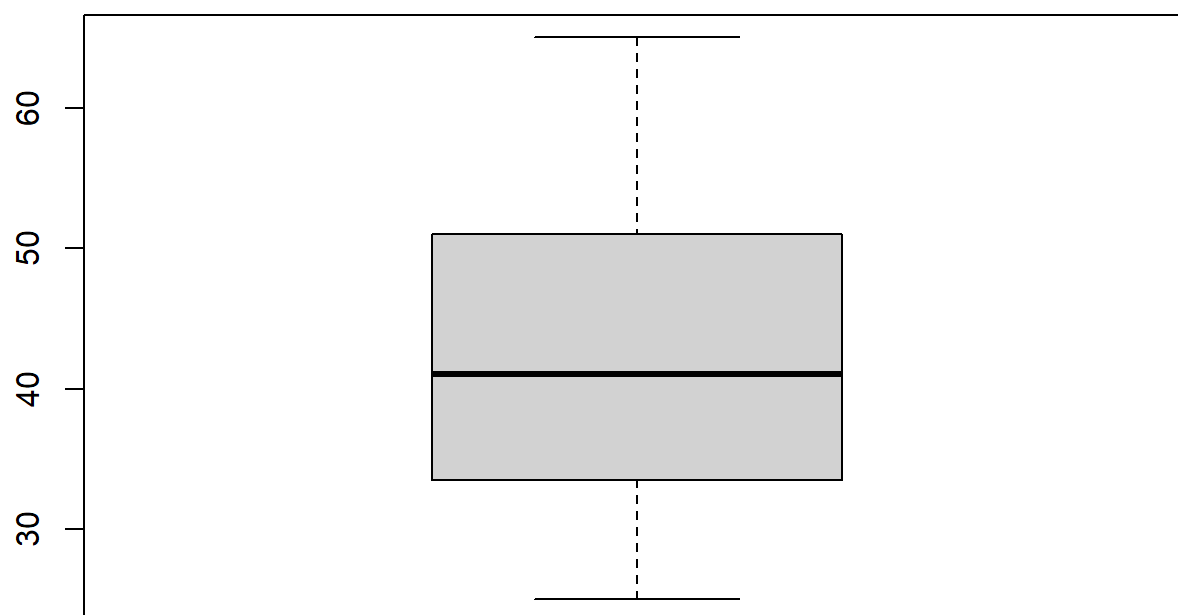


```
hist(medical_data$Glucose)
```

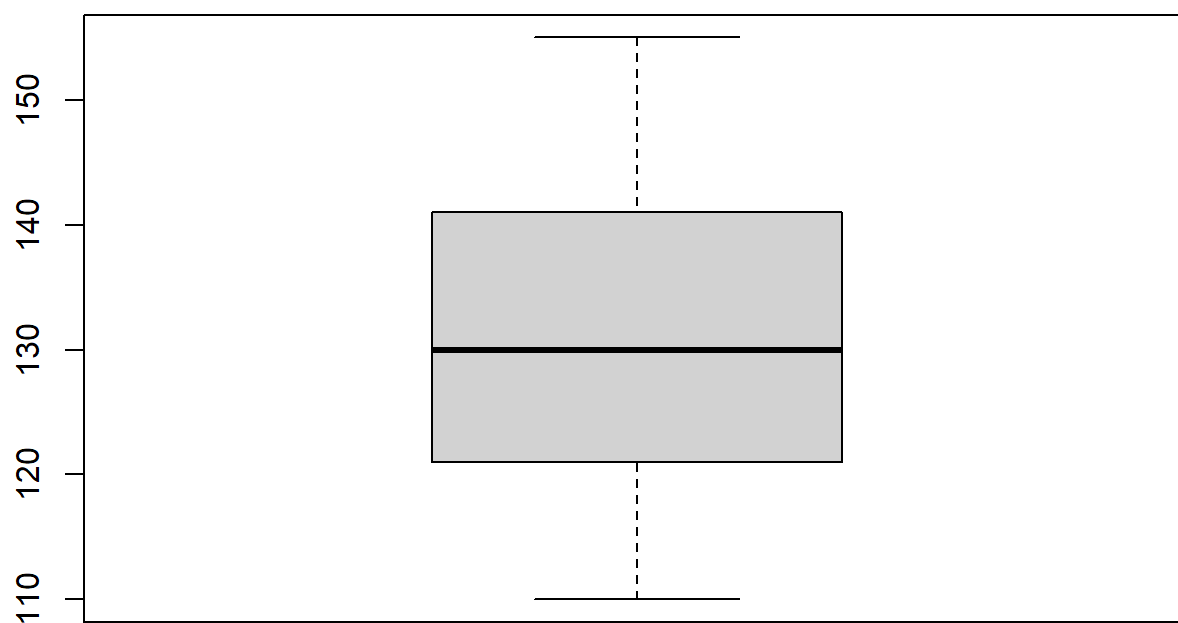
Histogram of medical_data\$Glucose



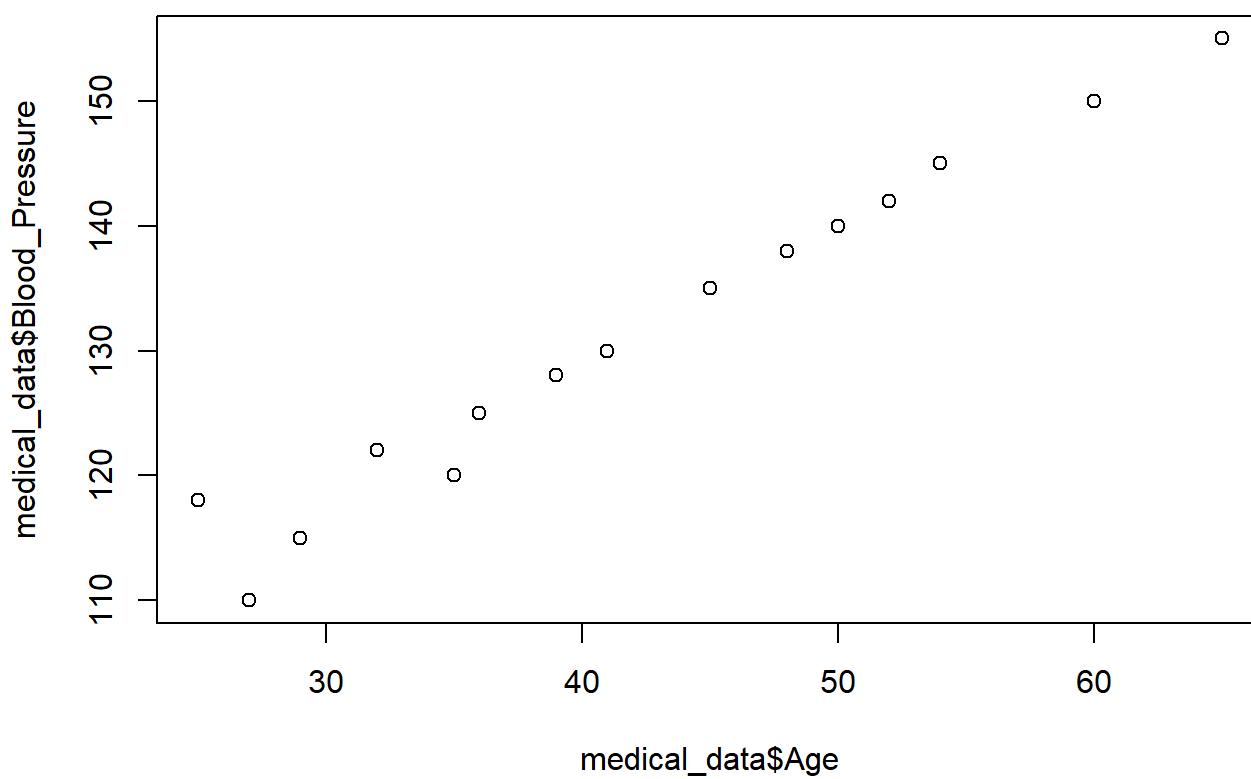
```
boxplot(medical_data$Age)
```



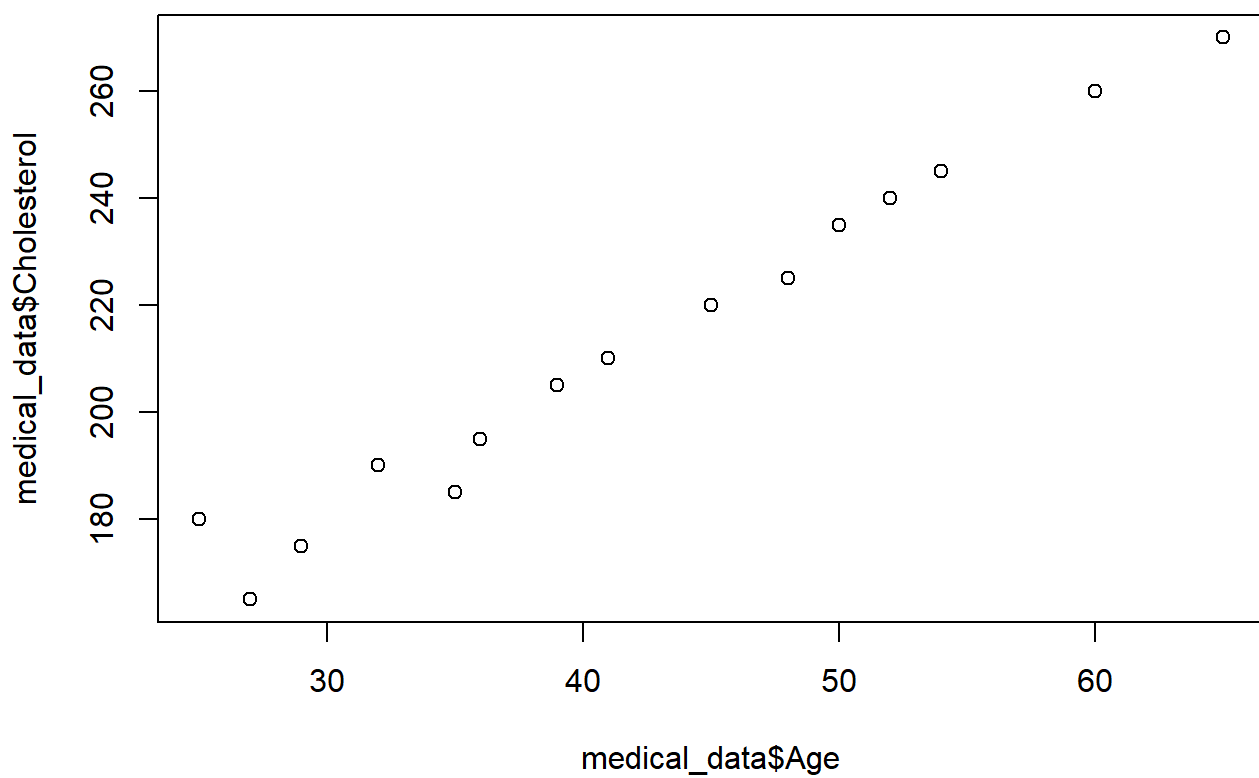
```
boxplot(medical_data$Blood_Pressure)
```



```
plot(medical_data$Age, medical_data$Blood_Pressure)
```



```
plot(medical_data$Age, medical_data$Cholesterol)
```



```
barplot(table(medical_data$Gender, medical_data$Health_Status), beside=TRUE, legend.text=TRUE)
```

